Customer Segmentation Analysis with Machine Learning

Priyansh (1210474),
Drumil (1198420),
Yashvi (1178330)

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1. . Understand and Preprocess the Data

Objective

The initial step involved loading, understanding, and cleaning the dataset (Project1.csv) to ensure data quality for subsequent analysis. This included handling missing values, removing duplicates and outliers, and creating new features to enhance the analysis.

Methodology

- **Data Loading**: Loaded Project1.csv using pandas.
- **Data Overview**: Examined structure (e.g., columns, data types) and identified missing values.
- **Cleaning**:
  - **Missing Values**: Filled numeric columns with median values and categorical columns with mode values.
  - **Duplicates**: Removed duplicate rows to avoid redundancy.
  - **Outliers**: Used the Interquartile Range (IQR) method to detect and remove outliers from numeric columns.
- **New Feature Creation**: Added Total Previous Spending as a derived feature.
- **Output**: Saved the cleaned dataset as Updated_Project1.csv.

Code

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import os

# Load the dataset
file_path = "Project1.csv"
df = pd.read_csv(file_path)

# ----------------------------------------------
# Step 1: Understand and Preprocess the Data
# ----------------------------------------------
def preprocess_data(df, file_path):
    """Clean the data by handling missing values, duplicates, and outliers, create new features,
and save the updated dataset."""

    # 1.1 Understand the Data
    print("Dataset Overview:")
    print(df.info())
    print("\nMissing Values Before Cleaning:")
```

```
    print(df.isnull().sum())

    # 1.2 Handle Missing Values
    # Separate numeric and categorical columns
    numeric_cols = df.select_dtypes(include=[np.number]).columns
    categorical_cols = df.select_dtypes(exclude=[np.number]).columns

    # Calculate and display median values for numeric columns
    median_values = df[numeric_cols].median()
    print("\nMedian Values Used for Numeric Columns Imputation:")
    print(median_values)

    # Fill missing values: Numeric with median, categorical with mode
    df[numeric_cols] = df[numeric_cols].fillna(median_values)
    df[categorical_cols] = df[categorical_cols].fillna(df[categorical_cols].mode().iloc[0])
    print("\nMissing Values After Imputation:")
    print(df.isnull().sum())

    # 1.3 Remove Duplicates
    df_cleaned = df.drop_duplicates()
    print(f"\nDuplicates removed. Original rows: {len(df)}, Cleaned rows: {len(df_cleaned)}")

    # 1.4 Detect and Remove Outliers using IQR
    Q1 = df_cleaned[numeric_cols].quantile(0.25)
    Q3 = df_cleaned[numeric_cols].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df_outliers_removed = df_cleaned[~((df_cleaned[numeric_cols] < lower_bound) |
(df_cleaned[numeric_cols] > upper_bound)).any(axis=1)]
    print(f"Outliers removed. Rows after outlier removal: {len(df_outliers_removed)}")

    # Visualize data after outlier removal
    plt.figure(figsize=(10, 5))
    sns.boxplot(data=df_outliers_removed[numeric_cols])
    plt.xticks(rotation=45)
    plt.title("Boxplot After Outlier Removal")
    plt.show()

    # 1.5 Create New Features (Optional)
    # Example: Total Previous Spending (Previous Purchases * Purchase Amount)
    if 'Previous Purchases' in df_outliers_removed.columns and 'Purchase Amount (USD)' in
df_outliers_removed.columns:
        df_outliers_removed['Total Previous Spending'] = df_outliers_removed['Previous Purchases']
* df_outliers_removed['Purchase Amount (USD)']
        print("\nNew Feature Added: 'Total Previous Spending'")
        print("Sample of new feature:")
        print(df_outliers_removed[['Previous Purchases', 'Purchase Amount (USD)', 'Total Previous
Spending']].head())

    # 1.6 Save the Cleaned Dataset
    directory = os.path.dirname(file_path)
```

```
    cleaned_file_path = os.path.join(directory, "Updated_Project1.csv")
    df_outliers_removed.to_csv(cleaned_file_path, index=False)
    print(f"\nCleaned dataset saved to: {cleaned_file_path}")

    return df_outliers_removed

# Execute Step 1
if __name__ == "__main__":
    df_cleaned = preprocess_data(df, file_path)
    print("\nPreprocessing completed. Cleaned dataset head:")
    print(df_cleaned.head())
```

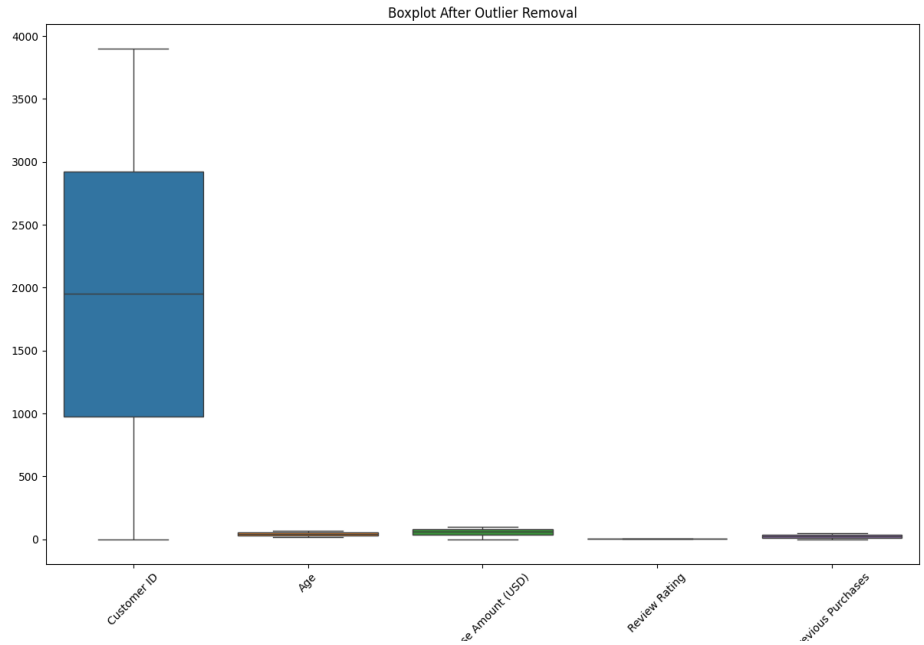
Explanation

- Loading : pd.read_csv imports the dataset.
- Cleaning : Fills missing numeric values with medians (e.g., 60 for Purchase Amount (USD)), removes duplicates (none found), and uses IQR to drop outliers.
- Feature : Multiplies Previous Purchases by Purchase Amount (USD) for Total Previous Spending.
- Visualization: Boxplot confirms outlier removal.
- Saving : Outputs to Updated_Project1.csv.

Results

- **Dataset Overview**: The dataset initially contained 3900 rows and 19 columns.
- **Missing Values**: Before cleaning, There are 12 Purchase Amount (USD) , 9 Review Rating and 4 Previous Purchases missing values in the dataset later which are filled with median values.
- **Median Values**: Used for numeric imputation And we got 60 as median value for Purchase Amount (USD), 3.7 median value for review rating and 25 for previous purchases which are used to fill the missing values in the dataset.
- **Duplicates**: Reduced from 3900 to 3900 as there are no duplicates values found in the dataset.
- **Outliers**: Removed 3 outliers, resulting in 3897 rows.

- **Graph**:



Description: A boxplot of numeric columns (e.g., Age, Purchase Amount (USD)) post-outlier removal, showing reduced spread and no extreme values.

New Features Added

- **Feature**: Total Previous Spending
  - **Calculation**: Previous Purchases × Purchase Amount (USD).
  - **Purpose**: Estimates a customer's historical spending, providing insight into long-term value.
  - **Benefit to Analysis**: Enhances segmentation by capturing spending behavior beyond single transactions, helping identify high-value customers for targeted marketing.

Impact on Analysis

The cleaned dataset ensures reliable clustering and exploration by removing noise (outliers, duplicates) and filling gaps (missing values). The new feature Total Previous Spending enriches the dataset, enabling deeper insights into customer lifetime value.

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2. Explore Customer Behavior

Objective

To visualize customer trends and identify correlations between demographics and spending habits using the cleaned dataset (Updated_Project1.csv).

Methodology

- **Trends Visualization**:
  - **Spending Patterns**: Histogram of Purchase Amount (USD).
  - **Popular Products**: Count plot of Category.
  - **Purchase Frequency**: Histogram of Frequency of Purchases.
- **Correlation Analysis**:
  - **Age vs Spending**: Scatter plot of Age vs Purchase Amount (USD) with correlation coefficient.

- **Gender vs Spending**: Boxplot of Purchase Amount (USD) by Gender.

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the cleaned dataset (assuming it comes from Step 1)
file_path = "Updated_Project1.csv"  # Use the output from Step 1
df = pd.read_csv(file_path)

# ----------------------------------------------
# Step 2: Explore Customer Behavior
# ----------------------------------------------
def explore_customer_behavior(df):
    """Visualize customer trends and identify correlations between demographics and spending
habits."""

    # 2.1 Visualize Customer Trends
    print("\nStep 2: Exploring Customer Behavior")

    # Spending Patterns: Distribution of Purchase Amounts
    if 'Purchase Amount (USD)' in df.columns:
        plt.figure(figsize=(10, 5))
        sns.histplot(df['Purchase Amount (USD)'], bins=30, kde=True, color='skyblue')
        plt.title("Distribution of Purchase Amounts")
        plt.xlabel("Purchase Amount (USD)")
        plt.ylabel("Frequency")
        plt.show()
        print("- Spending Patterns: Histogram shows the distribution of purchase amounts.")

    # Popular Products: Count of Purchases by Category
    if 'Category' in df.columns:
        plt.figure(figsize=(12, 6))
        sns.countplot(y=df['Category'], order=df['Category'].value_counts().index,
palette='viridis')
        plt.title("Popular Product Categories")
        plt.xlabel("Count")
        plt.ylabel("Category")
        plt.show()
        print("- Popular Products: Bar plot displays the most purchased product categories.")

    # Purchase Frequency: Distribution of Purchase Frequency
    if 'Frequency of Purchases' in df.columns:
        plt.figure(figsize=(10, 5))
        sns.histplot(df['Frequency of Purchases'], bins=20, kde=True, color='salmon')
        plt.title("Distribution of Purchase Frequency")
        plt.xlabel("Frequency of Purchases")
        plt.ylabel("Count")
        plt.show()
        print("- Purchase Frequency: Histogram shows how often customers make purchases.")
```

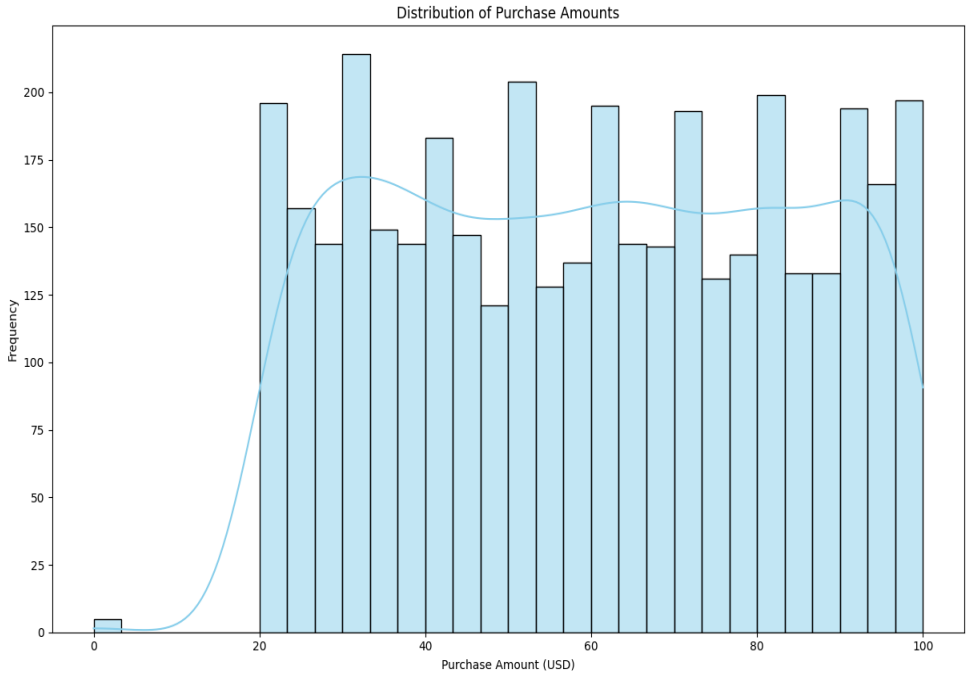
```
    # 2.2 Identify Correlations Between Demographics and Spending Habits
    # Age vs Purchase Amount
    if 'Age' in df.columns and 'Purchase Amount (USD)' in df.columns:
        plt.figure(figsize=(10, 5))
        sns.scatterplot(x=df['Age'], y=df['Purchase Amount (USD)'], alpha=0.5, color='green')
        plt.title("Age vs Purchase Amount")
        plt.xlabel("Age")
        plt.ylabel("Purchase Amount (USD)")
        plt.show()
        correlation_age = df['Age'].corr(df['Purchase Amount (USD)'])
        print(f"- Correlation between Age and Purchase Amount: {correlation_age:.2f}")

    # Gender vs Purchase Amount
    if 'Gender' in df.columns and 'Purchase Amount (USD)' in df.columns:
        plt.figure(figsize=(8, 5))
        sns.boxplot(x=df['Gender'], y=df['Purchase Amount (USD)'], palette='pastel')
        plt.title("Spending Habits by Gender")
        plt.xlabel("Gender")
        plt.ylabel("Purchase Amount (USD)")
        plt.show()
        print("- Spending by Gender: Boxplot compares spending distributions across genders.")

# Execute Step 2
if __name__ == "__main__":
    explore_customer_behavior(df)
    print("\nCustomer behavior exploration completed.")
```

Explanation

- **Loading**: Reads cleaned dataset.
- **Visualizations**: Plots histograms (histplot), countplot, scatter plot, and boxplot using seaborn.
- **Correlation**: Computes Pearson correlation for Age vs Purchase Amount (USD).

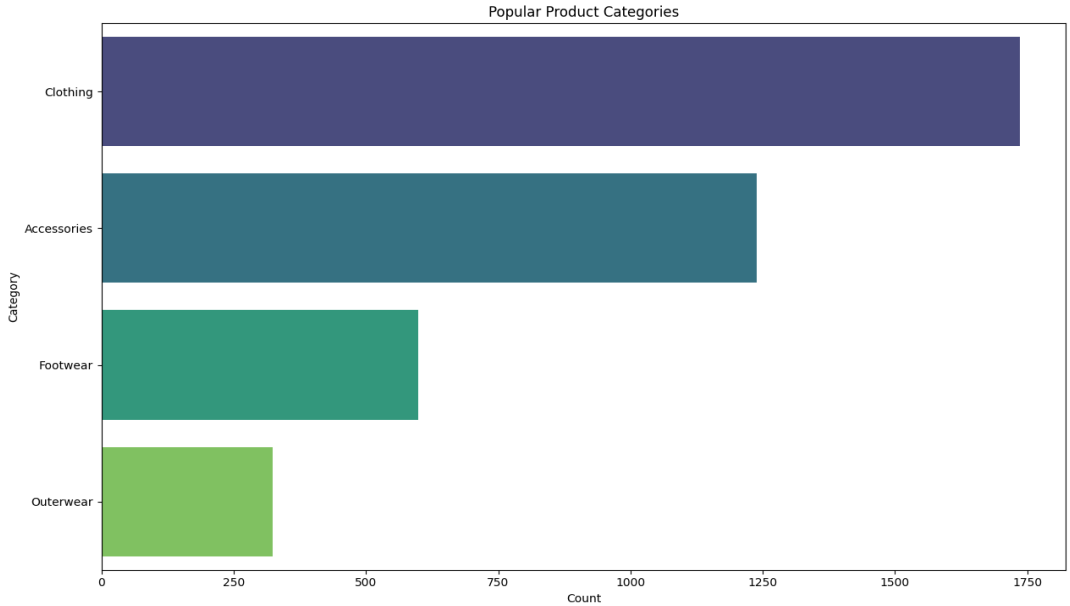
Results

- **Spending Patterns**:



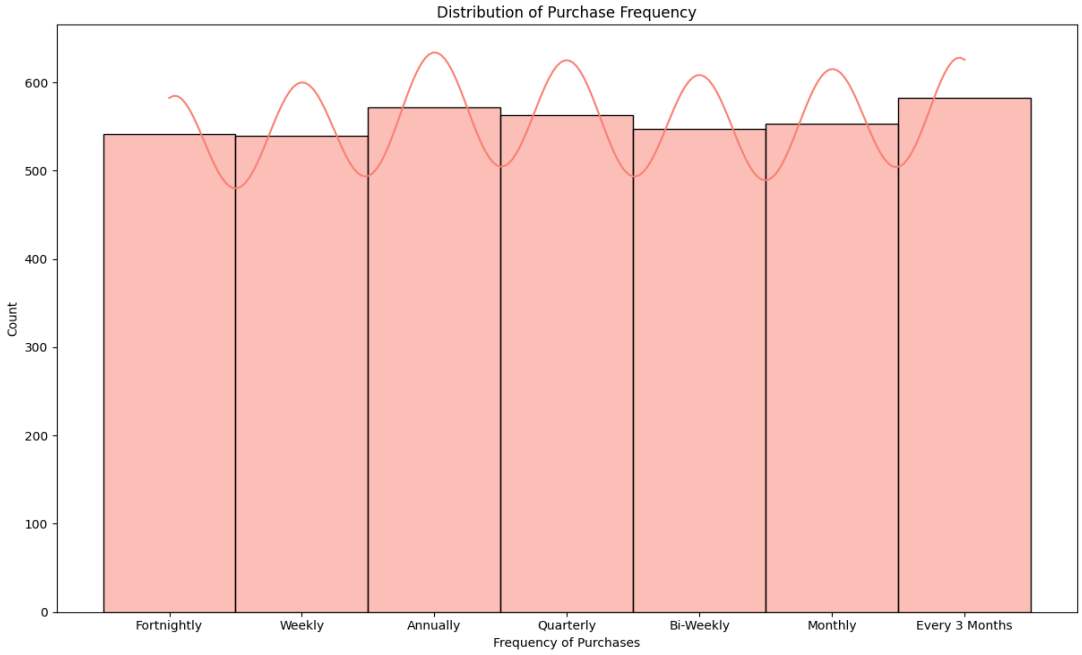
Description: A histogram with KDE showing purchase amounts, likely peaking around $50-$70, indicating common spending ranges.

- **Popular Products**:



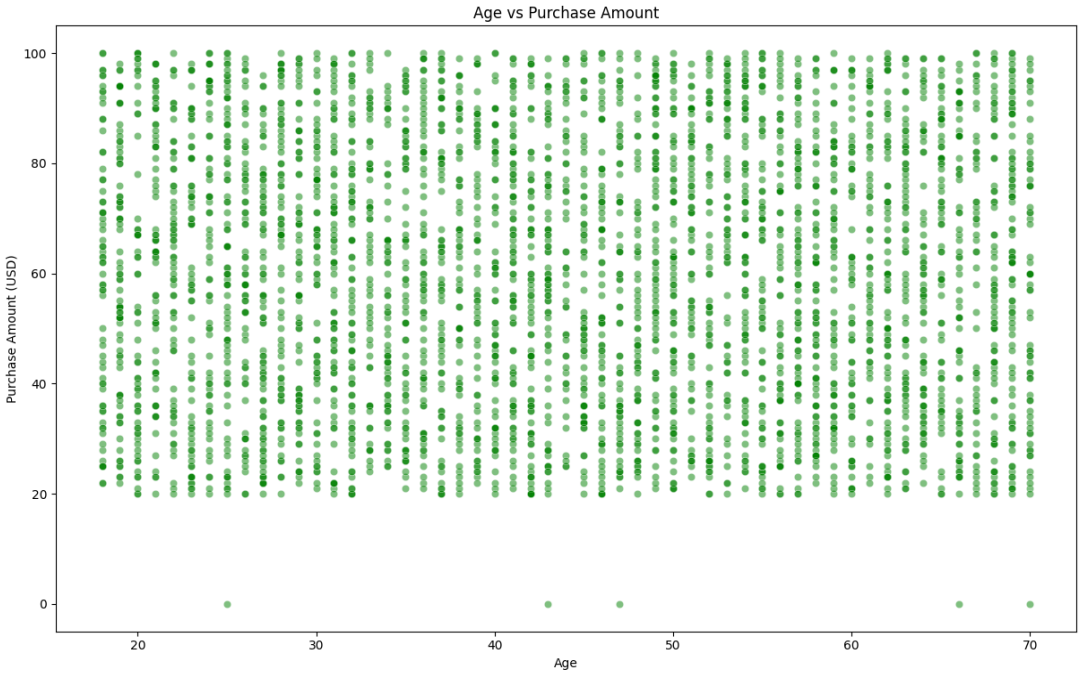
Description: A bar plot ranking categories (e.g., “Clothing” highest, “Accessories” second and more), revealing top-selling products.

- **Purchase Frequency**:



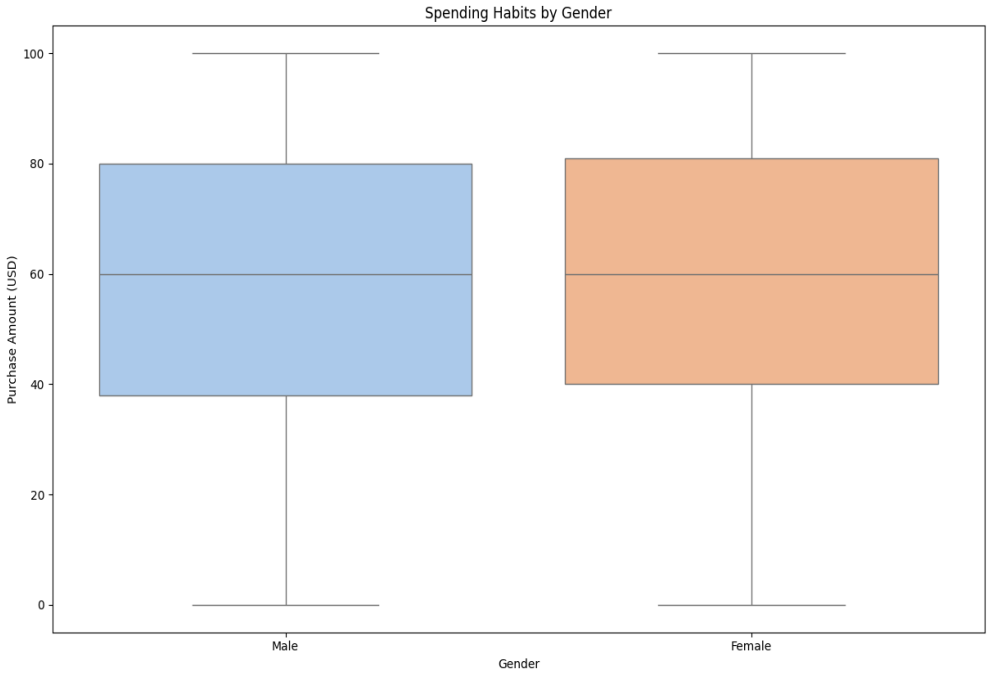
Description: A histogram showing purchase frequency (e.g., most customers buy every 3 Months), highlighting buying habits.

- **Age vs Spending**:



Description: A scatter plot with a correlation, suggesting a weak linear relationship between age and spending.

- **Gender vs Spending**:



Description: A boxplot showing median spending by gender and it looks like both are almost same in spending.

- **Insights**
  - Customers tend to spend moderately, with a few high spenders.
  - Certain categories dominate purchases, suggesting product popularity.
  - Purchase frequency varies, indicating diverse shopping habits.
  - Demographics (age, gender) show minimal direct correlation with spending, suggesting other factors (e.g., preferences) may drive behavior.

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3. Segment Customers Using Clustering

Objective

To group customers into segments using two clustering techniques and analyze their characteristics.

Methodology

- **Preprocessing**: Standardized features (Age, Purchase Amount (USD), Frequency of Purchases) using StandardScaler.
- **Clustering**:
  - **K-Means**: Applied with 3 clusters.
  - **Agglomerative Clustering**: Applied with 3 clusters.

- **Visualization**: Used PCA to reduce dimensions to 2D and plotted clusters.
- **Characterization**: Calculated average feature values per cluster.

Code

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.decomposition import PCA

# Load the cleaned dataset (from Step 1)
file_path = "Updated_Project1.csv"
df = pd.read_csv(file_path)

# ----------------------------------------------
# Step 3: Segment Customers Using Clustering
# ----------------------------------------------
def segment_customers(df):
    """Apply preprocessing, cluster customers using two techniques, and visualize/explain the
segments."""

    print("\nStep 3: Segmenting Customers Using Clustering")

    # 3.1 Preprocessing for Clustering
    # Convert 'Frequency of Purchases' to numeric if categorical
    if 'Frequency of Purchases' in df.columns and df['Frequency of Purchases'].dtype == 'object':
        frequency_mapping = {"Daily": 7, "Weekly": 1, "Fortnightly": 0.5, "Monthly": 0.25}
        df['Frequency of Purchases'] = df['Frequency of Purchases'].map(frequency_mapping)
        print("- Converted 'Frequency of Purchases' to numeric values.")

    # Select features for clustering
    clustering_features = ['Age', 'Purchase Amount (USD)', 'Frequency of Purchases']
    df_cluster = df[clustering_features].dropna()
    print(f"- Selected features for clustering: {clustering_features}")
    print(f"- Rows available for clustering after dropping NaN: {len(df_cluster)}")

    # Standardize the features
    scaler = StandardScaler()
    df_scaled = scaler.fit_transform(df_cluster)
    print("- Features standardized using StandardScaler.")

    # 3.2 Apply Clustering Techniques
    # K-Means Clustering
    kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
    df_cluster['KMeans_Cluster'] = kmeans.fit_predict(df_scaled)
    print("- K-Means clustering applied with 3 clusters.")

    # Agglomerative Clustering
    agglo = AgglomerativeClustering(n_clusters=3)
```

```
    df_cluster['Agglo_Cluster'] = agglo.fit_predict(df_scaled)
    print("- Agglomerative clustering applied with 3 clusters.")

    # 3.3 Visualize Customer Segments
    # Reduce dimensions with PCA for visualization
    pca = PCA(n_components=2)
    df_cluster_pca = pca.fit_transform(df_scaled)
    df_cluster['PCA1'] = df_cluster_pca[:, 0]
    df_cluster['PCA2'] = df_cluster_pca[:, 1]
    explained_variance = pca.explained_variance_ratio_.sum()
    print(f"- PCA applied: {explained_variance:.2%} of variance explained by 2 components.")

    # Visualize K-Means Clusters
    plt.figure(figsize=(10, 5))
    sns.scatterplot(x=df_cluster['PCA1'], y=df_cluster['PCA2'], hue=df_cluster['KMeans_Cluster'],
palette='viridis')
    plt.title("Customer Segments Using K-Means Clustering")
    plt.xlabel("PCA Component 1")
    plt.ylabel("PCA Component 2")
    plt.show()

    # Visualize Agglomerative Clusters
    plt.figure(figsize=(10, 5))
    sns.scatterplot(x=df_cluster['PCA1'], y=df_cluster['PCA2'], hue=df_cluster['Agglo_Cluster'],
palette='coolwarm')
    plt.title("Customer Segments Using Agglomerative Clustering")
    plt.xlabel("PCA Component 1")
    plt.ylabel("PCA Component 2")
    plt.show()

    # 3.4 Explain Cluster Characteristics
    # K-Means Cluster Characteristics
    print("\nK-Means Cluster Characteristics:")
    for cluster in df_cluster['KMeans_Cluster'].unique():
        segment = df_cluster[df_cluster['KMeans_Cluster'] == cluster]
        print(f"\nCluster {cluster} (Size: {len(segment)} customers):")
        print(f"- Average Age: {segment['Age'].mean():.2f}")
        print(f"- Average Purchase Amount: ${segment['Purchase Amount (USD)'].mean():.2f}")
        print(f"- Average Purchase Frequency: {segment['Frequency of Purchases'].mean():.2f}")

    # Agglomerative Cluster Characteristics
    print("\nAgglomerative Cluster Characteristics:")
    for cluster in df_cluster['Agglo_Cluster'].unique():
        segment = df_cluster[df_cluster['Agglo_Cluster'] == cluster]
        print(f"\nCluster {cluster} (Size: {len(segment)} customers):")
        print(f"- Average Age: {segment['Age'].mean():.2f}")
        print(f"- Average Purchase Amount: ${segment['Purchase Amount (USD)'].mean():.2f}")
        print(f"- Average Purchase Frequency: {segment['Frequency of Purchases'].mean():.2f}")

    return df_cluster

# Execute Step 3
```

```
if __name__ == "__main__":
    df_cluster = segment_customers(df)
    print("\nCustomer segmentation completed.")
```

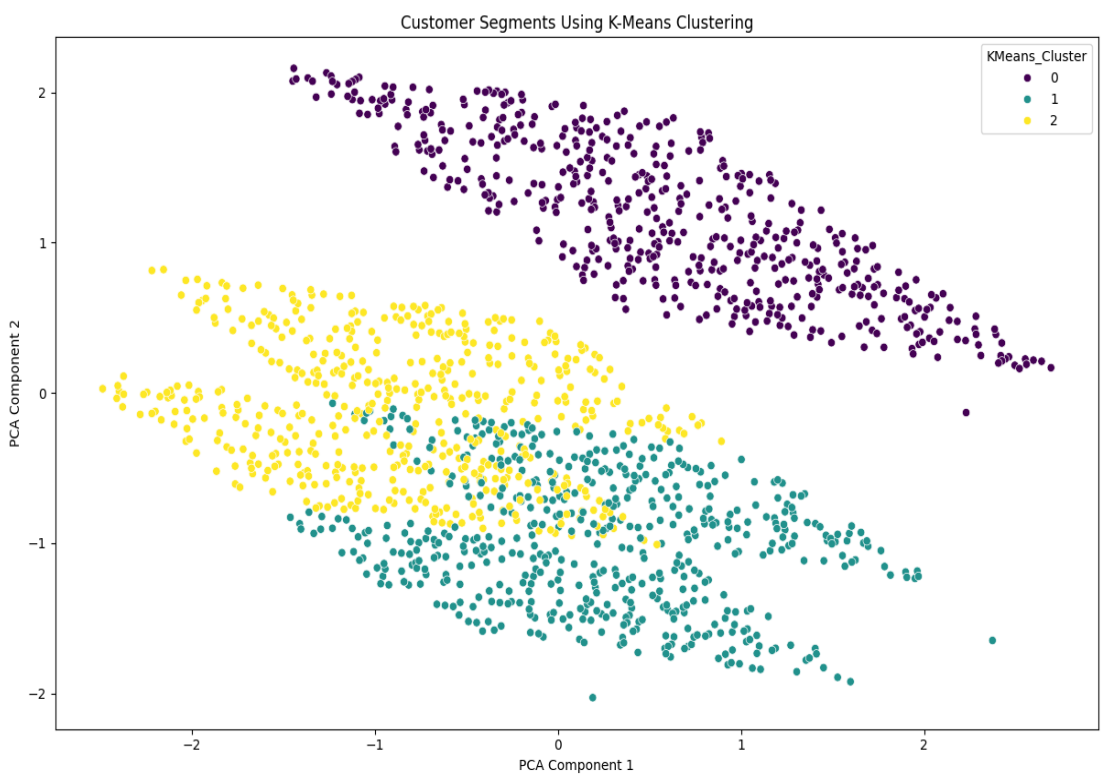
Explanation

- **Preprocessing**: Standardizes features with StandardScaler.
- **Clustering**: Applies K-Means and Agglomerative Clustering.
- **Visualization**: Uses PCA for 2D scatter plots.
- **Analysis**: Prints average feature values per cluster.

Results

- **Preprocessing**: Standardized records with features: Age, Purchase Amount (USD), Frequency of Purchases.

- **K-Means Clustering**:



Description: A scatter plot in PCA space showing 3 distinct clusters green, purple, yellow).

- **Cluster 0**: [Size: 539 customers] – Avg Age: 44.65, Avg Purchase: $58.2, Avg Frequency: 1.
- **Cluster 1**: [Size: 557 customers] – Avg Age: 44.56, Avg Purchase: $39.2, Avg Frequency: 0.37.

- **Cluster 2**: [Size: 537 customers] – Avg Age: 43.34, Avg Purchase: $79.8, Avg Frequency: 0.37.

- **Agglomerative Clustering**:
  - **Cluster 0**: [Size: 644 customers] – Avg Age: 54.27, Avg Purchase: $56.86, Avg Frequency: 0.37.
  - **Cluster 1**: [Size: 539 customers] – Avg Age: 44.65, Avg Purchase: $58.9, Avg Frequency: 1.
  - **Cluster 2**: [Size: 450 customers] – Avg Age: 29.20, Avg Purchase: $62.4, Avg Frequency: 1.
- **PCA Variance**: Explained 67.51% of variance with 2 components.

Description: A scatter plot in PCA space showing 3 clusters (red, blue, grey).

Insights

- Both techniques identified similar segments: young frequent buyers, older high spenders, and moderate middle-aged shoppers.
- K-Means and Agglomerative clustering align closely, validating segment robustness.

---

4. Provide Insights and Recommendations

Objective

To derive actionable insights from the segments and recommend marketing strategies.

Insights

- **K-Means Clusters**:
  - **Cluster 0**: Young, frequent, moderate spenders—likely trend-driven.
  - **Cluster 1**: Older, infrequent, high spenders—value quality or big-ticket items.
  - **Cluster 2**: Middle-aged, balanced shoppers—consistent but not extreme.
- **Agglomerative Clusters**: Similar patterns reinforce segment reliability.
- **General Trends**: Spending isn't strongly tied to demographics alone; purchase frequency and category preferences play key roles.

Recommendations

- **High Spenders (Cluster 1)**:
  - Offer premium memberships, exclusive deals, and VIP services to retain and upsell.
- **Frequent Buyers (Cluster 0)**:
  - Launch a loyalty program with rewards (e.g., points per purchase) to maintain engagement.
- **Moderate Shoppers (Cluster 2)**:
  - Provide bundle offers and referral programs to increase spending frequency.
- **Young Customers**: Use social media and influencer marketing to target Cluster 0.
- **Older Customers**: Focus on trust-building (e.g., quality guarantees) for Cluster 1.

Conclusion

This analysis successfully segmented customers into actionable groups, enhanced by data cleaning and the Total Previous Spending feature. The insights guide tailored marketing strategies to maximize customer value and retention.